Rua 使用手册

v1.1.0

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目录

1 mkinfo

`rua mkinfo` 用于生成目标平台的构建指令。该指令包含了众多常用的 make 变量，用于方便地定制构建行为。例如：通过传入 `-6/--ipv6` 选项，令构建目标中包含 `ipv6` 字样；通过传入 `-w/--webui` 选项，将在指令中的 `NOTBUILDWEBUI` 变量设置为 `0`（带 WebUI）。

1.1 用法

用户可通过 `rua mkinfo -h` 查看帮助信息：

```
❯ rua mkinfo -h
Get all matched makeinfos for product

Usage: rua mkinfo [OPTIONS] <NAME-OR-TARGET>

Arguments:
  <NAME-OR-TARGET>  Product name such as A1000, or build target (with --target
switch on) such as a-dnv, View as a product name by default. Regex is also
supported when using as a product name, e.g. 'X\d+80'

Options:
  -4, --ipv4                          Build with only IPv4 enabled
  -6, --ipv6                          Build with IPv6 enabled
  -g, --coverage                      Run coverage
  -c, --coverity                      Run coverity
  -d, --debug                         Build in debug mode (default is release
mode)
      --format <FORMAT>               Output format for makeinfos, defaults to
list [default: list] [possible values: csv, json, list, tsv]
  -p, --password                      Build with shell password enabled
  -w, --webui                         Build with WebUI enabled
  -s, --image-server <IMAGE-SERVER>   Server to upload the output image to
[possible values: b, s]
      --nostrip <BINARY>              Binaries that get out of strip processing
      --by-target                     Treat the positional arg as a build target
other than a product name
  -h, --help                          Print help

Examples:
  rua mkinfo A1000      # Makeinfo for A1000 without extra features
  rua mkinfo -6 A1000   # Makeinfo for A1000 with IPv6 enabled
  rua mkinfo -6w 'X\d+' # Makeinfos for X-series products with IPv6 and WebUI
enabled using regex pattern
  rua mkinfo --target a-dnv  # Makeinfos for a-dnv target
```

1.2 示例

1. `rua mkinfo -6 A1000`：生成 A1000 平台的构建指令，启用 IPv6 支持

```
bzhao in 🌐 build18-bj in ~/repos/REL_R11
❯ rua mkinfo -6 A1000
1 matched info:
================================================================================================================================================================================================
Product   : SG-6000-A1000
Model     : HS_TYPE_SG6000_A1000
Family    : HS_PRODUCT_FAMILY_FIREWALL
Platform  : HS_A1000
Target    : a-dnv-ipv6
Directory : products/ngfw_as/
Command   : hsdocker7 "make -C products/ngfw_as/ -j8 a-dnv-ipv6 ISBUILDRELEASE=1 NOTBUILDUNIWEBUI=1 HS_SHELL_PASSWORD=0 HS_BUILD_COVERAGE=0 HS_BUILD_COVERITY=0 OS_IMA
GE_FTP_IP=10.200.6.10 IMG_NAME=SG6000-RELR11-ADNV-V6-r0215-bzhao >build.log 2>&1"
================================================================================================================================================================================================
Run make command under the project root, i.e. "/home/bzhao/repos/REL_R11"
```

2. `rua mkinfo -w A1000`：生成 A1000 平台的构建指令，启用 WebUI 支持

```
bzhao in 🌐 build18-bj in ~/repos/REL_R11
❯ rua mkinfo -w A1000
1 matched info:
================================================================================================================================================================================================
Product   : SG-6000-A1000
Model     : HS_TYPE_SG6000_A1000
Family    : HS_PRODUCT_FAMILY_FIREWALL
Platform  : HS_A1000
Target    : a-dnv
Directory : products/ngfw_as/
Command   : hsdocker7 "make -C products/ngfw_as/ -j8 a-dnv ISBUILDRELEASE=1 NOTBUILDUNIWEBUI=0 HS_SHELL_PASSWORD=0 HS_BUILD_COVERAGE=0 HS_BUILD_COVERITY=0 OS_IMAGE_FT
P_IP=10.200.6.10 IMG_NAME=SG6000-RELR11-ADNV-r0215-bzhao >build.log 2>&1"
================================================================================================================================================================================================
Run make command under the project root, i.e. "/home/bzhao/repos/REL_R11"
```

3. `rua mkinfo -6w A1000`：生成 A1000 平台的构建指令，启用 IPv6 支持以及 WebUI 支持

```
bzhao in 🌐 build18-bj in ~/repos/REL_R11
❯ rua mkinfo -6w A1000
1 matched info:
================================================================================================================================================================================================
Product   : SG-6000-A1000
Model     : HS_TYPE_SG6000_A1000
Family    : HS_PRODUCT_FAMILY_FIREWALL
Platform  : HS_A1000
Target    : a-dnv-ipv6
Directory : products/ngfw_as/
Command   : hsdocker7 "make -C products/ngfw_as/ -j8 a-dnv-ipv6 ISBUILDRELEASE=1 NOTBUILDUNIWEBUI=0 HS_SHELL_PASSWORD=0 HS_BUILD_COVERAGE=0 HS_BUILD_COVERITY=0 OS_IMA
GE_FTP_IP=10.200.6.10 IMG_NAME=SG6000-RELR11-ADNV-V6-r0215-bzhao >build.log 2>&1"
================================================================================================================================================================================================
Run make command under the project root, i.e. "/home/bzhao/repos/REL_R11"
```

4. `rua mkinfo -ss A1000`：生成 A1000 平台的构建指令，上传到 10.200.6.10 服务器，即苏州服务器

```
bzhao in 🌐 build18-bj in ~/repos/REL_R11
❯ rua mkinfo -s s A1000
1 matched info:
================================================================================================================================================================================================
Product   : SG-6000-A1000
Model     : HS_TYPE_SG6000_A1000
Family    : HS_PRODUCT_FAMILY_FIREWALL
Platform  : HS_A1000
Target    : a-dnv
Directory : products/ngfw_as/
Command   : hsdocker7 "make -C products/ngfw_as/ -j8 a-dnv ISBUILDRELEASE=1 NOTBUILDUNIWEBUI=1 HS_SHELL_PASSWORD=0 HS_BUILD_COVERAGE=0 HS_BUILD_COVERITY=0 OS_IMAGE_FT
P_IP=10.200.6.10 IMG_NAME=SG6000-RELR11-ADNV-r0215-bzhao >build.log 2>&1"
================================================================================================================================================================================================
Run make command under the project root, i.e. "/home/bzhao/repos/REL_R11"
```

5. `rua mkinfo --format json A1000`：生成 A1000 平台的构建指令，输出格式指定为 JSON 格式，适合脚本使用

```
bzhao in 🌐 build18-bj in ~/repos/MX_MAIN
❯ rua mkinfo --format json A1000
[
  {
    "Command": "hsdocker7 \"make -C products/ngfw_as/ -j8 a-dnv ISBUILDRELEASE=1 NOTBUILDUNIWEBUI=1 HS_SHELL_PASSWORD=0 HS_BUILD_COVERAGE=0 HS_BU
ILD_COVERITY=0 OS_IMAGE_FTP_IP=10.200.6.10 IMG_NAME=SG6000-MXMAIN-ADNV-r0331-bzhao >build.log 2>&1\"",
    "Directory": "products/ngfw_as/",
    "Family": "HS_PRODUCT_FAMILY_FIREWALL",
    "Model": "HS_TYPE_SG6000_A1000",
    "OEMID": "HS_OEMID_BJHS",
    "Platform": "HS_A1000",
    "Product": "SG-6000-A1000",
    "Target": "a-dnv"
  }
]
```

2 compdb

rua compdb 包含了众多子命令，分别用于生成、删除和管理编译数据库。

编译数据库能够为 C/C++语言服务器（Language Server，LS）提供各编译单元的编译指令，如 include 路径、编译宏，使之能够理解每个文件的编译方式，从而在代码代码库中正确地行走。

编译数据库的格式为 JSON 格式，文件名为 compile_commands.json，存放在当前工作目录下。该文件是一个数组，每个元素对应一个编译单元的编译指令。

编译数据库的每个元素包含了以下几个字段：

- directory：编译单元所在的目录
- command：编译单元的编译指令
- file：编译单元的文件路径

一般来说，编译数据库是分构建目标的，如 a-dnv-ipv6 对应有一个编译数据库，a-dnv 对应有另一个编译数据库。对于后者而言，IPV6 宏所包裹的代码不会被解析，LS 视之为无效代码。

Rua 工具为每条命令和参数添加了充分的注释，当存在疑惑时请使用 -h 或 --help 参数查看帮助信息。例如，工具顶层命令帮助信息可通过在 shell 下执行 rua compdb -h 查看：

```
❯ rua compdb -h
Manipulate compilation database

Usage: rua compdb <COMMAND>

Commands:
  gen     Generate a JSON compilation database (JCDB) for the given target
[aliases: generate]
  add     Archive the currently used compilation database into store as a new
generation [aliases: ark, archive]
  del     Delete compilation database generation(s) from store [aliases: delete,
rm, remove]
  ls      List all compilation database generations in store [aliases: list]
  use     Select a compilation database generation from store to use
  name    Name a compilation database generation
  remark  Remark a compilation database generation
  help    Print this message or the help of the given subcommand(s)

Options:
  -h, --help  Print help
```

2.1 生成编译数据库(Gen)

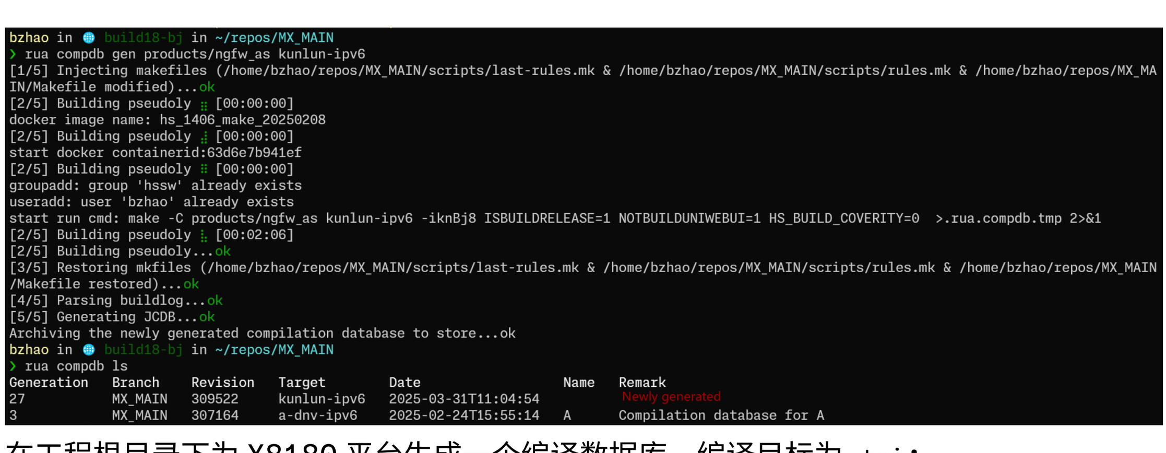
```
❯ rua compdb gen -h
Generate a JSON compilation database (JCDB) for the given target

Usage: rua compdb gen [OPTIONS] <PATH> <TARGET>

Arguments:
  <PATH>    Path for the target where platform-specific makefiles reside, such
as 'products/vfw'
  <TARGET>  Target to build, such as 'a-dnv'

Options:
  -D, --define <KEY=VAL>
          Define a variable which will be passed to the underlying make command
  -e, --engine <ENGINE>
          Engine for generating compilation database (defaults to built-in)
[possible values: built-in, intercept-build, bear]
  -b, --bear-path <BEAR>
          Path to the bear binary (defaults to /devel/sw/bear/bin/bear)
  -i, --intercept-build-path <INTERCEPT-BUILD>
          Path to the intercept-build binary (defaults to /devel/sw/llvm/bin/
intercept-build)
  -h, --help
          Print help (see more with '--help')

Examples:
  rua compdb gen products/ngfw_as a-dnv                          # For A1000/A2000...
  rua compdb gen products/ngfw_as a-dnv-ipv6                     # For A1000/A2000...
with IPv6 support
  rua compdb gen -e intercept-build products/ngfw_as a-dnv # For A1000/A2000...
using intercept-build
  rua compdb gen . a-dnv                                         # For A1000/A2000...
under submod dir
  rua compdb gen -e bear . a-dnv                                 # For A1000/A2000...
under submod dir using bear
  run compdb gen -e intercept-build . a-dnv                      # For A1000/A2000...
under submod dir using intercept-build

Caution:
  Some files are modified while running in built-in mode which is the default
  and faster:
  1. When running under project root dir:
     - scripts/last-rules.mk
     - scripts/rules.mk
     - Makefile
  2. When running under submod dir:
     - scripts/last-rules.mk
     - scripts/rules.mk
  These files may be left dirty if compdb process aborted unexpectedly. You
  could manually restore them by execute:
  svn revert Makefile scripts/last-rules.mk scripts/rules.mk
```

参数解析：

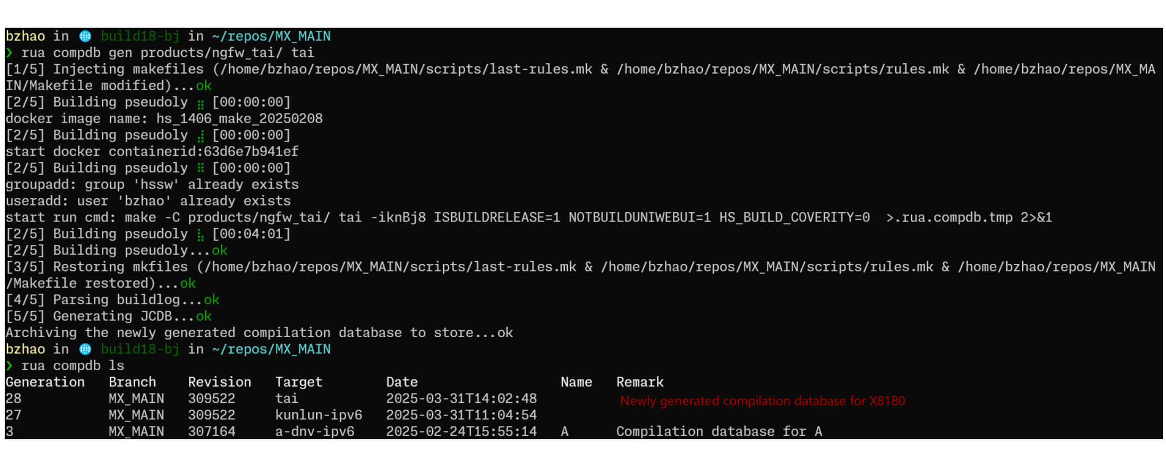
- <PATH>：构建路径，例如 A1000/A2000 平台的构建路径为 products/ngfw_as，K6580 平台的构建路径为 products/ngfw_ak or …
- <TARGET>：构建目标，例如 A1000/A2000 等平台的编译目标为 a-dnv，K6580 平台的编译目标位 hygon，X8180 平台的编译目标位 tai
- -e/--engine：可选，引擎。候选值为 built-in、intercept-build、bear，默认值为 built-in
- -D/--define：可选，变量定义。将会传递给底层的 make 命令
- -b/--bear-path：可选，指定 bear 的路径，默认值为 /devel/sw/bear/bin/bear
- -i/--intercept-build-path：可选，指定 intercept-build 的路径，默认值为 /devel/sw/llvm/bin/intercept-build

2.1.1 使用默认方法(built-in)生成编译数据库

1. 在工程根目录下生成一个编译目标为 kunlun-ipv6 的编译数据库：



2. 在工程根目录下为 X8180 平台生成一个编译数据库，编译目标为 tai：



2.1.2 使用 intercept-build 方法生成编译数据库

intercept-build 是 LLVM 工具集合所提供的一个工具，能够拦截编译命令并生成编译数据库。

由于该工具的工作原理为拦截每次触发编译器编译的指令，因此不适合增量编译场景。增量编译场景下，编译器只会编译那些有变更的文件，而不会编译所有文件。这样一来，intercept-build 就无法捕获到所有的编译指令。 因此，建议在全量编译时使用该工具。

2.1.3 FQA

1. rua compdb gen 执行失败怎么办？
   命令执行失败的原因有以下几个：
   - 无执行权限，建议使用 chmod +x <RUA-PATH> 添加可执行权限
   - 执行目录不正确，建议在工程根目录下执行
   - Makefile 状态不正确，简易检查 “scripts/rules.mk”、“scripts/last-rules.mk”、“Makefile” 三个文件是否被修改过。若有修改，建议执行 svn revert 恢复
   - 磁盘满了，建议检查磁盘空间是否足够。该类错误不但会导致编译数据库生成失败，还可能造成 “scripts/rules.mk”、“scripts/last-rules.mk”、“Makefile” 三个文件被修改
2. 无法生成第三方库的编译数据库怎么办？
   rua compdb gen 默认使用 built-in 方法，该方法只生成我们的内部代码，暨通过 “scripts/last-rules.mk” 中一条用于编译目的的 recipe 来编译的文件。
   若想单独生成第三方库的编译数据库，可使用 bear 或 intercept-build 方法。
   若想使用 bear 或 intercept-build 方法生成整个工程的编译数据库，须在一个 clean 过的的分支下进行（从而进行全量编译），或者编译一个当前未被 ccache 缓存其编译信息的编译目标，否则会失败。

2.2 归档编译数据库(Add)

归档功能适合将其他工具生成的编译数据库归档到 store 中，以便管理。其用法如下：

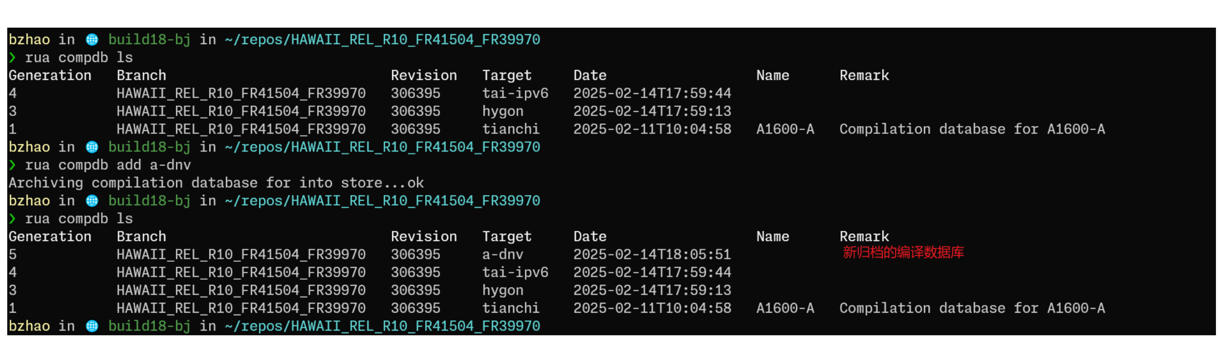
```
❯ rua compdb add --help
Archive the currently used compilation database into store as a new generation

Usage: rua compdb add [OPTIONS] <TARGET>

Arguments:
  <TARGET>  Target specified for the compilation database

Options:
  -r, --revision <REVISION>
          Revision for compilation database (defaults to current repo revision)
  -f, --compilation-database <COMPILATION-DATABASE>
          Use this compilation database other than the default
(compile_commands.json)
  -h, --help
          Print help

Examples:
    rua compdb add hygon-ipv6 # Archive compilation database for hygon-ipv6
    rua compdb add --revision 307164 hygon # Archive compilation database for
hygon with a revision provided
```

参数解析：

- <TARGET>：必填，用于指示编译数据库所对应的构建目标，如 a-dnv、hygon
- -r/--revision：可选，用于指示编译数据库所对应的代码版本，默认使用当前工作目录的代码版本
- -f/--compilation-database：可选，用于手动指定编译数据库路径，默认使用当前工作目录下的 compile_commands.json

例如：

将当前工作目录下的 compile_commands.json 添加到 store 中：

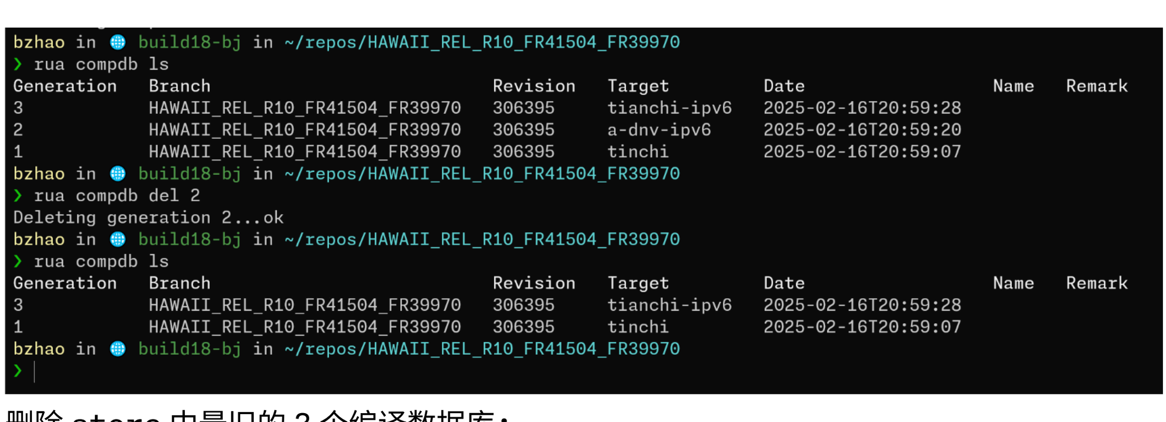


2.3 删除编译数据库(Del)

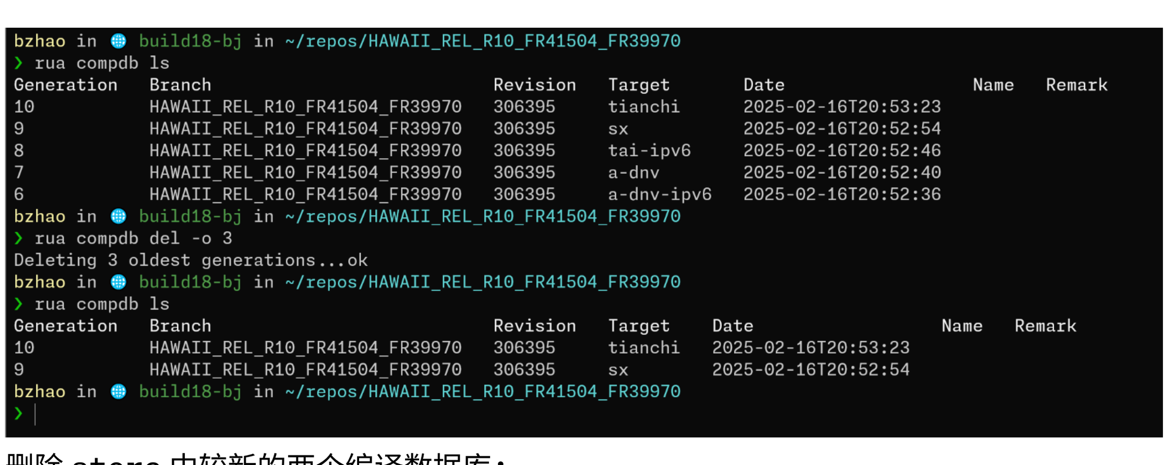
```
❯ rua compdb del -h
Delete compilation database generation(s) from store

Usage: rua compdb del [OPTIONS] [GENERATION-ID]

Arguments:
  [GENERATION-ID]  Generation to delete

Options:
  -a, --all      Remove all generations
  -n, --new <N>  Remove N newest generations
  -o, --old <N>  Remove N oldest generations
  -h, --help     Print help
```

例如：

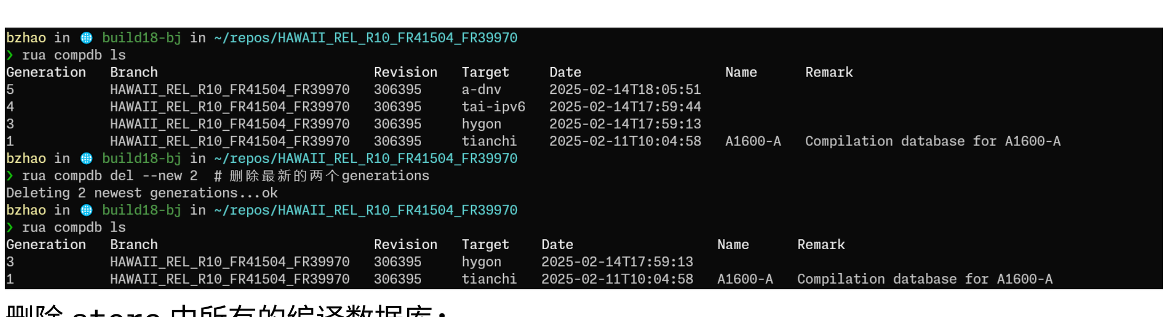
1. 删除 store 中的 Generation 2：



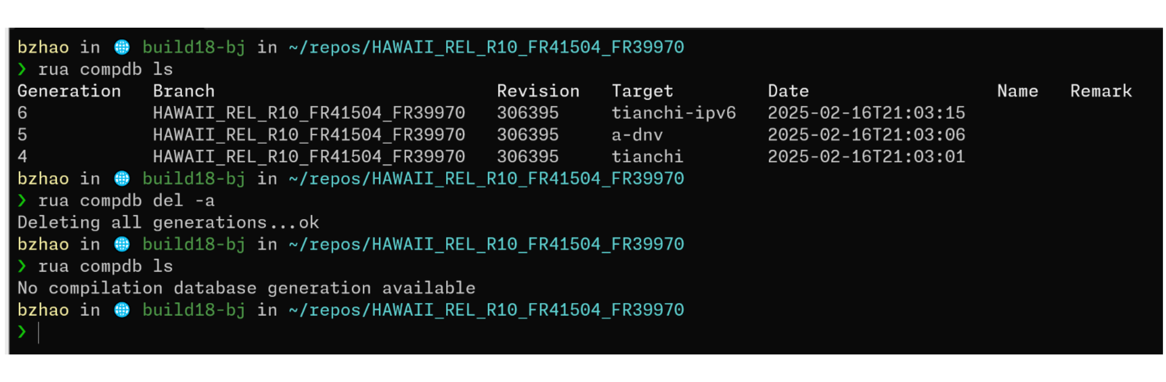
2. 删除 store 中最旧的 3 个编译数据库：



3. 删除 store 中较新的两个编译数据库：



4. 删除 store 中所有的编译数据库：



2.4 列出编译数据库(Ls)

列出当前工作目录下所有的编译数据库。

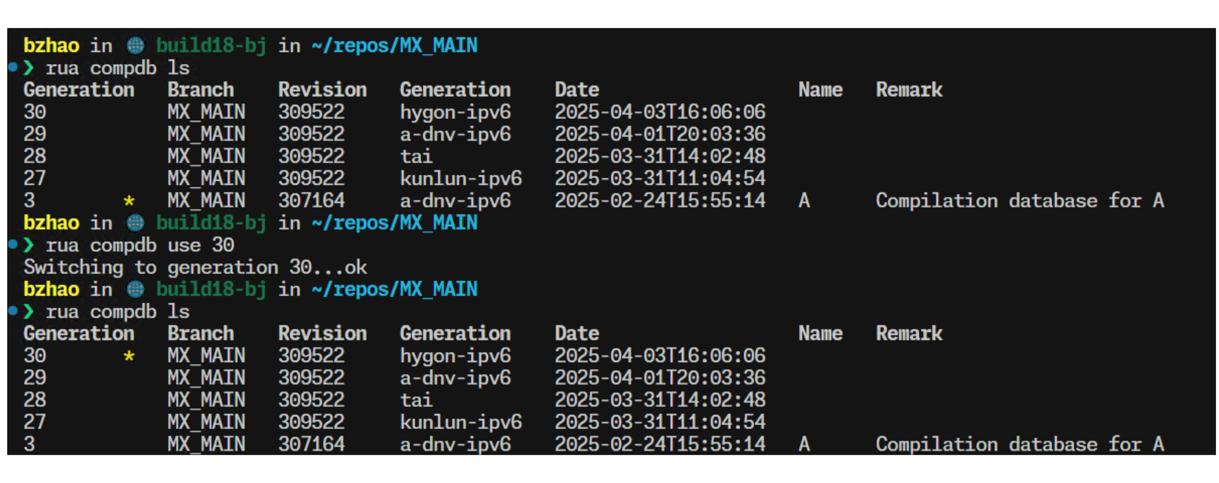
```
❯ rua compdb ls -h
List all compilation database generations in store

Usage: rua compdb ls

Options:
  -h, --help  Print help
```

每个表项都有一个唯一的 Generation ID，且关联 4 个重要属性和 2 个可选属性：

- Revision：代码版本
- Target：构建目标
- Date：生成日期
- Name：可选，编译数据库的名字
- Remark：可选，编译数据库的备注

当一个 generation 后面标有黄色的 * 时，表示它是它是当前使用项。gen/use 操作会使当前使用项发生改变，因此 * 的显示会发生改变。需要注意的是，为保证准确性，尽量使用 rua 来生成、切换数据库。

例如：



2.5 选择编译数据库(Use)

选中的编译数据库将会覆盖当前工作目录下的 compile_commands.json 文件。

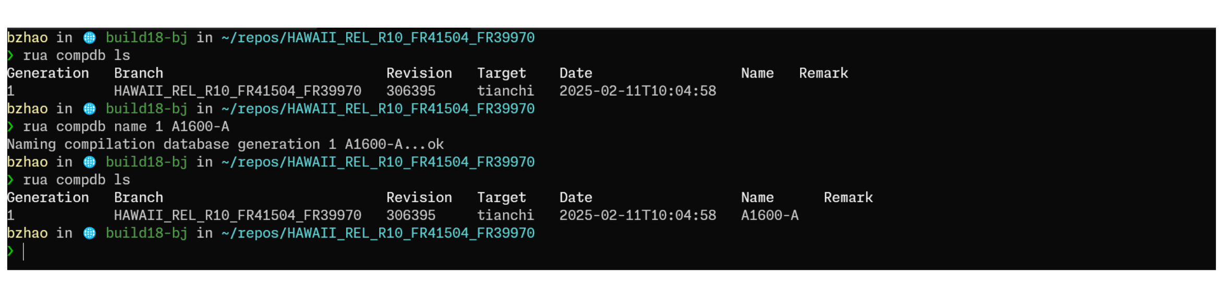
```
❯ rua compdb use -h
Select a compilation database generation from store to use

Usage: rua compdb use <GENERATION>

Arguments:
  <GENERATION>  Compilation database generation id

Options:
  -h, --help  Print help
```

2.6 命名编译数据库(Name)

```
❯ rua compdb name -h
Name a compilation database generation

Usage: rua compdb name <GENERATION> <NAME>

Arguments:
  <GENERATION>  The compilation database generation
  <NAME>        Name for the compilation database

Options:
  -h, --help  Print help
```

例如：

为 store 中的编译数据库 generation 1 添加一个名字：

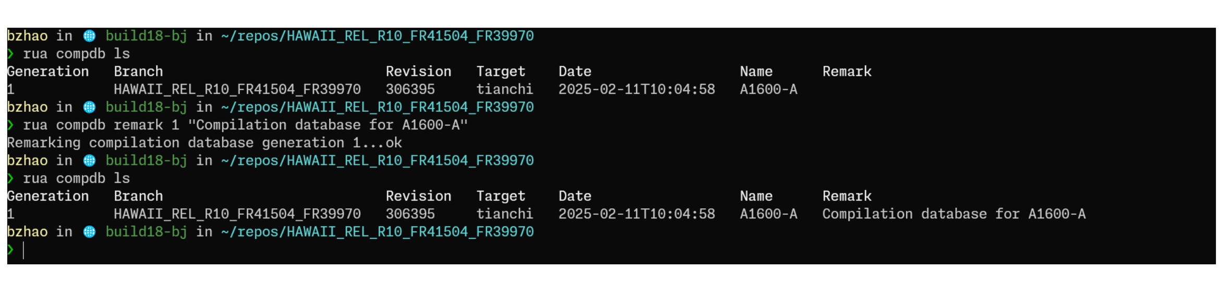


2.7 备注编译数据库(Remark)

```
❯ rua compdb remark -h
rua compdb remark <GENERATION-ID> <备注>
Remark a compilation database generation

Usage: rua compdb remark <GENERATION> <REMARK>

Arguments:
  <GENERATION>  The compilation database generation
  <REMARK>      Remark for the compilation database generation

Options:
  -h, --help  Print help
```

例如：

为 store 中的编译数据库 Generation 1 添加备注



3 showcc

显示某个文件的编译指令，依赖于编译数据库。

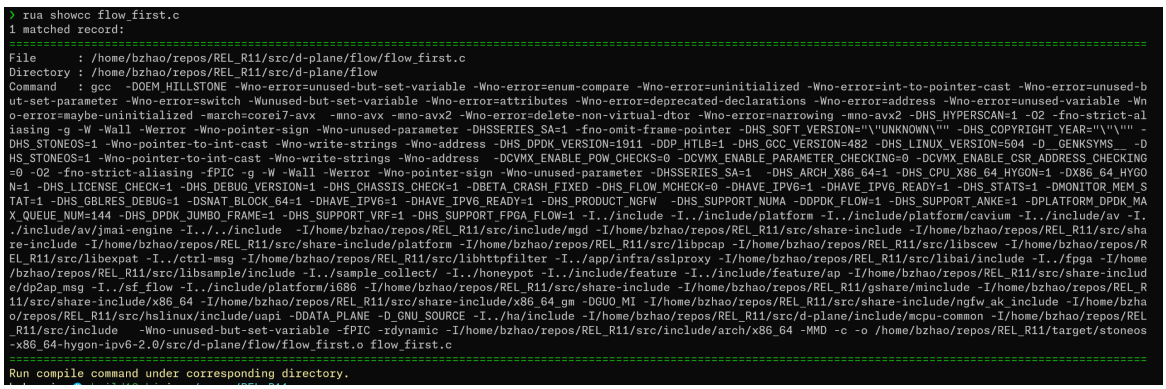
3.1 用法

用户可通过 `rua showcc -h` 查看帮助信息：

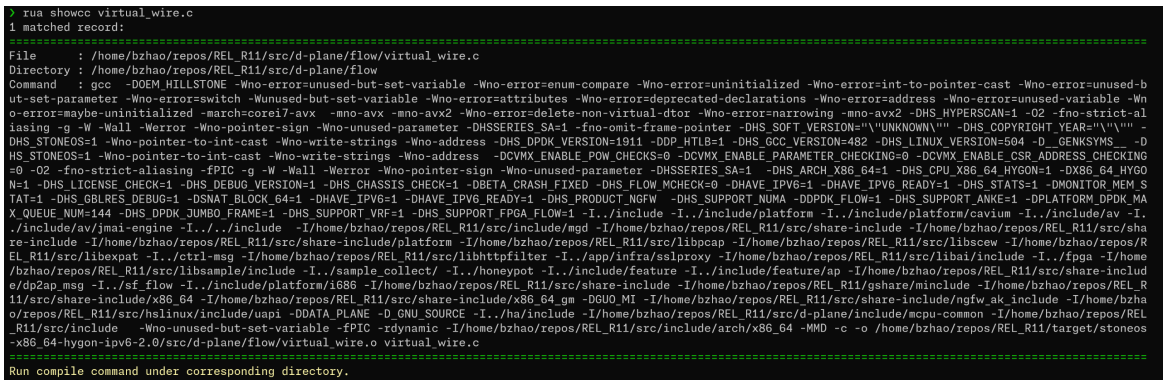
```
❯ rua showcc -h
Show all possible compile commands for filename (based on compilation database)

Usage: rua showcc [OPTIONS] <SOURCE-FILE>

Arguments:
  <SOURCE-FILE>  Source file name for which to fetch all the available compile
commands

Options:
  -c, --compdb <COMPDB>  Compilation database (defaults to file
"compile_commands.json" in the current directory)
  -h, --help             Print help
```

3.2 示例

1. 显示 `flow_first.c` 文件的编译指令：



2. 显示 `virtual_wire.c` 的编译指令：



4 review

该子命令与 autoreview-cops 工具相似，参数更符合直觉。

4.1 用法

```
● [podman] ❯ rua review -h
Start a new review request or refresh the existing one if review-id provided

Usage: rua review [OPTIONS] --bug <BUG> [FILE]...

Arguments:
  [FILE]...  Files to be reviewed

Options:
  -n, --bug <BUG>                      Bug id for this review request (required)
  -r, --review-id <REVIEW-ID>          Existing review id
  -d, --diff-file <DIFF-FILE>          Diff file to be used
  -u, --reviewers <REVIEWERS>          Reviewers
  -b, --branch <BRANCH>                Branch name for this commit
  -p, --repo <REPO>                    Repository name
  -s, --revision <REVISION>            Revision to be used
  -t, --template-file <TEMPLATE-FILE>  Use customized template file (please
ensure it can run through svn commit hooks)
  -h, --help                           Print help
```

5 perfan

rua perfan 用于对 profiling text 进行指令地址映射。

5.1 用法

```
Extensively map instructions to file locations (inline expanded)

Usage: rua perfan [OPTIONS] <FILE>

Arguments:
  <FILE>  Profiling text generated by perfan

Options:
  -o, --format <FORMAT>  Output format [default: table] [possible values: json,
table]
  -e, --elf <ELF>        Binary files used for addresses resolving [aliases:
exe, executable]
  -h, --help             Print help
```

5.2 示例

1. 在 MX_MAIN 分支下，使用 rua perfan 命令解析 profiling 文本中属于 d-plane 的地址：

```
● ❯ cat A3600.profile.txt
  Samples |      Source code & Disassembly of d-plane for cycles (7209 samples, percent: local period)
 ---------------------------------------------------------------------------------------------------
       1 :   609ba0:       push   %rbp
       2 :   609ba1:       mov    %rsp,%rbp
       4 :   609ba6:       push   %r14
       1 :   609ba8:       push   %r13
       1 :   609baa:       push   %r12
       2 :   609bac:       push   %rbx
       2 :   609bad:       sub    $0xa8,%rsp
     121 :   609bb4:       rdtsc
       1 :   609bb9:       mov    %eax,%eax
       4 :   609bbb:       lea    0x4a055be(%rip),%rdx        # 500f180 <dp_glb_vars>
       1 :   609bc6:       lea    0x4a05553(%rip),%rbx        # 500f120 <flow_q_head>
       4 :   609bcd:       mov    %rsi,-0xb8(%rbp)
      19 :   609bd4:       or     %rax,-0xb8(%rbp)
       1 :   609be2:       mov    (%rax),%rax
       1 :   609be5:       mov    %rax,-0xc0(%rbp)
       4 :   609bec:       lea    0x55e3add(%rip),%rax        # 5bed6d0 <dp_current_core_id>
       4 :   609bf3:       mov    (%rax),%eax
       4 :   609bf7:       imul   $0x61c0,%rcx,%rcx
       1 :   609bfe:       lea    0x6d0(%rcx,%rdx,1),%rsi
       3 :   609c06:       mov    (%rbx),%rdx
       2 :   609c09:       mov    %rsi,-0xd0(%rbp)
       4 :   609c1d:       prefetcht0 (%rdx)
       5 :   609c26:       lea    0x55e3b13(%rip),%rax        # 5bed740 <dp_glb_vars_local>
       3 :   609c2d:       mov    $0x200,%esi
       1 :   609c35:       mov    0x460(%rax),%r14d
       2 :   609c3c:       mov    $0x20,%eax
       1 :   609c44:       cmovle %esi,%eax
       2 :   609c57:       test   %r13d,%r13d
       1 :   609c5a:       jg     60a3b5 <hs_flow_process_packets+0x815>
       4 :   609c60:       lea    0x55e3a71(%rip),%rax        # 5bed6d8 <dp_debug_buf_data_len>
       2 :   609c67:       mov    (%rax),%r12d
       2 :   609c6a:       test   %r12d,%r12d
       4 :   609c6d:       jne    60a3cd <hs_flow_process_packets+0x82d>
       2 :   609c73:       lea    0x55e3a4a(%rip),%rax        # 5bed6c4 <debug_ctrl_flag>
```

原始 A3600 profiling 文本

```
● ❯ rua perfan -e ./bin/obj-emulator-a-dnv-ipv6-2.0/d-plane ./A3600.profile.txt
                                                                |#samples:62246|#daemons:28|#funcs:386|#lines:4604|

◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆|d-plane|percentage:94.3563%|#samples:58733/62246|#funcs:139/386|#lines:3965/4604|◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆

Percentage         Count   Address   Instruction                    Location
  11.5815%   7209/62246                [d-plane]
   0.0016%      1/62246    609ba0 push   %rbp                       hs_flow_process_packets@flow_main.c:1115
   0.0032%      2/62246    609ba1 mov    %rsp,%rbp                  hs_flow_process_packets@flow_main.c:1115
   0.0064%      4/62246    609ba6 push   %r14                       hs_flow_process_packets@flow_main.c:1115
   0.0016%      1/62246    609ba8 push   %r13                       hs_flow_process_packets@flow_main.c:1115
   0.0016%      1/62246    609baa push   %r12                       hs_flow_process_packets@flow_main.c:1115
   0.0032%      2/62246    609bac push   %rbx                       hs_flow_process_packets@flow_main.c:1115
   0.0032%      2/62246    609bad sub    $0xa8,%rsp                 hs_flow_process_packets@flow_main.c:1115
   0.1944%    121/62246    609bb4 rdtsc                             hs_flow_process_packets@flow_main.c:1118->time_count_reg_get@i686_core_drv.h:
169
   0.0016%      1/62246    609bb9 mov    %eax,%eax                  hs_flow_process_packets@flow_main.c:1118->time_count_reg_get@i686_core_drv.h:
170
   0.0064%      4/62246    609bbb lea    0x4a055be(%rip),%rdx       hs_flow_process_packets@flow_main.c:1122
   0.0016%      1/62246    609bc6 lea    0x4a05553(%rip),%rbx       hs_flow_process_packets@flow_main.c:1124
   0.0064%      4/62246    609bcd mov    %rsi,-0xb8(%rbp)           hs_flow_process_packets@flow_main.c:1118->time_count_reg_get@i686_core_drv.h:
170
   0.0305%     19/62246    609bd4 or     %rax,-0xb8(%rbp)           hs_flow_process_packets@flow_main.c:1118->time_count_reg_get@i686_core_drv.h:
170
   0.0016%      1/62246    609be2 mov    (%rax),%rax                hs_flow_process_packets@flow_main.c:1120
   0.0016%      1/62246    609be5 mov    %rax,-0xc0(%rbp)           hs_flow_process_packets@flow_main.c:1120
   0.0064%      4/62246    609bec lea    0x55e3add(%rip),%rax       hs_flow_process_packets@flow_main.c:1121
   0.0064%      4/62246    609bf3 mov    (%rax),%eax                hs_flow_process_packets@flow_main.c:1121
```

A3600 profiling 文本经 rua perfan 解析后

2. 传入多个 elf 参数，解析多个二进制的地址：

```
rua perfan -e ./bin/obj-emulator-a-dnv-ipv6-2.0/d-plane -e ./bin/obj-
emulator-a-dnv-ipv6-2.0/netd A3600.profile.txt
```

注意： 当传入非 C 语言（包括 C++）生成的二进制中，符号是混淆过的，函数名可能比较奇怪，这是正常现象。结果中提供有文件名和行号，凭此信息能够准确地定位代码行。

5.3 输出格式

```
                                                                |#samples:62246|#daemons:28|#funcs:386|#lines:4604|  profiling 概况
46 pending changes
◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆|d-plane|percentage:94.3563%|#samples:58733/62246|#funcs:139/386|#lines:3965/4604|◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆
Percentage  采样占比   Count   Address   Instruction                    采样占比   Location
  11.5815%   7209/62246                [d-plane]   所属elf
   0.0016%      1/62246    609ba0 push   %rbp                       hs_flow_process_packets@flow_main.c:1115
   0.0032%      2/62246    609ba1 mov    %rsp,%rbp                  hs_flow_process_packets@flow_main.c:1115
   0.0064%      4/62246    609ba6 push   %r14                       hs_flow_process_packets@flow_main.c:1115
   0.0016%      1/62246    609ba8 push   %r13                       hs_flow_process_packets@flow_main.c:1115
   0.0016%      1/62246    609baa push   %r12                       hs_flow_process_packets@flow_main.c:1115   函数名@文件名:代码行
   0.0032%      2/62246    609bac push   %rbx                       hs_flow_process_packets@flow_main.c:1115   ->内敛函数名@文件名:代码行
   0.0032%      2/62246    609bad sub    $0xa8,%rsp                 hs_flow_process_packets@flow_main.c:1115
   0.1944%    121/62246    609bb4 rdtsc                             hs_flow_process_packets@flow_main.c:1118->time_count_reg_get@i686_core_drv.h:169
   0.0016%      1/62246    609bb9 mov    %eax,%eax                  hs_flow_process_packets@flow_main.c:1118->time_count_reg_get@i686_core_drv.h:170
   0.0064%      4/62246    609bbb lea    0x4a055be(%rip),%rdx       hs_flow_process_packets@flow_main.c:1122
   0.0016%      1/62246    609bc6 lea    0x4a05553(%rip),%rbx       hs_flow_process_packets@flow_main.c:1124
   0.0064%      4/62246    609bcd mov    %rsi,-0xb8(%rbp)           hs_flow_process_packets@flow_main.c:1118->time_count_reg_get@i686_core_drv.h:170
   0.0305%     19/62246    609bd4 or     %rax,-0xb8(%rbp)           hs_flow_process_packets@flow_main.c:1118->time_count_reg_get@i686_core_drv.h:170
   0.0016%      1/62246    609be2 mov    (%rax),%rax                hs_flow_process_packets@flow_main.c:1120
   0.0016%      1/62246    609be5 mov    %rax,-0xc0(%rbp)           hs_flow_process_packets@flow_main.c:1120
   0.0064%      4/62246    609bec lea    0x55e3add(%rip),%rax       hs_flow_process_packets@flow_main.c:1121
   0.0064%      4/62246    609bf3 mov    (%rax),%eax                hs_flow_process_packets@flow_main.c:1121
   0.0064%      4/62246    609bf7 imul   $0x61c0,%rcx,%rcx          hs_flow_process_packets@flow_main.c:1122
   0.0016%      1/62246    609bfe lea    0x6d0(%rcx,%rdx,1),%rsi    hs_flow_process_packets@flow_main.c:1122
   0.0048%      3/62246    609c06 mov    (%rbx),%rdx                hs_flow_process_packets@flow_main.c:1124
   0.0032%      2/62246    609c09 mov    %rsi,-0xd0(%rbp)           hs_flow_process_packets@flow_main.c:1122
   0.0064%      4/62246    609c1d prefetcht0 (%rdx)                 hs_flow_process_packets@flow_main.c:1124
   0.0080%      5/62246    609c26 lea    0x55e3b13(%rip),%rax       hs_flow_process_packets@flow_main.c:1128
   0.0048%      3/62246    609c2d mov    $0x200,%esi                hs_flow_process_packets@flow_main.c:1117
   0.0016%      1/62246    609c35 mov    0x460(%rax),%r14d          hs_flow_process_packets@flow_main.c:1117
   0.0032%      2/62246    609c3c mov    $0x20,%eax                 hs_flow_process_packets@flow_main.c:1117
   0.0016%      1/62246    609c44 cmovle %esi,%eax                  hs_flow_process_packets@flow_main.c:1117
   0.0032%      2/62246    609c57 test   %r13d,%r13d                hs_flow_process_packets@flow_main.c:1139
   0.0016%      1/62246    609c5a jg     60a3b5 <hs_flow_process_pack hs_flow_process_packets@flow_main.c:1139
   0.0064%      4/62246    609c60 lea    0x55e3a71(%rip),%rax       hs_flow_process_packets@flow_main.c:1142
   0.0032%      2/62246    609c67 mov    (%rax),%r12d               hs_flow_process_packets@flow_main.c:1142
   0.0032%      2/62246    609c6a test   %r12d,%r12d                hs_flow_process_packets@flow_main.c:1142
   0.0064%      4/62246    609c6d jne    60a3cd <hs_flow_process_pack hs_flow_process_packets@flow_main.c:1142
   0.0032%      2/62246    609c73 lea    0x55e3a4a(%rip),%rax       hs_flow_process_packets@flow_main.c:1143
   0.0016%      1/62246    609c7a xor    %r14d,%r14d                hs_flow_process_packets@flow_main.c:1138
   0.0048%      3/62246    609c7d mov    (%rbx),%r15                hs_flow_process_packets@flow_main.c:1159
```

Rua perfan 输出格式解析

5.4 工具比较

代码库中 tool/perf2func 具有相同的功能，本工具的优势在于：

- 速度更快，相比于 perf2func 提速 1500 倍左右
- 输出格式更友好，对内联函数有较好的展开处理，方便用户定位代码行

6 clean

清除工程根目录下的所有编译文件。该功能类似于最新 MX_MAIN 分支上（进行了 Makefile 优化）的 stoneos-clean。此前存在一些问题，本次完善了该功能。clean 会做以下三个工作：
1. 删除 target 文件夹
2. 删除 webui 文件夹（与分支同名）
3. 删除未受 SVN 管控的文件，改功能与 make stoneos-clean 相似。

Rua clean 允许开发者自己指定所要保留的文件（通过配置文件或命令行参数）。

6.1 用法

用户可以通过 rua init -h 查看帮助信息：

```
❯ rua clean -h
Clean build files (run under project root)

Usage: rua clean [OPTIONS] [ENTRY]...

Arguments:
  [ENTRY]...  Files or dirs to be cleaned ('target' is always included even if
not specified)

Options:
  -n, --ignore <FILE>  File or directory to be ignored while cleaning. You can
add multiple ignores by specifying this option multiple times
  -h, --help           Print help

Examples:
  rua clean  # Clean the entire project

Caution:
  All unversioned files will be REMOVED permanantly, including files created by
YOU but not
  added to SVN. Use it carefully!
```

6.2 示例

1. 清除工程根目录下的所有编译文件：

   ```
   rua clean .
   ```

   注意：该命令会删除当前目录下的所有编译文件，包括 webui 文件夹和 target 文件夹
2. 清除工程根目录下的所有编译文件，但保留文件夹 .rua、.cache 和文件 .ignore 和 compile_commands.json：

   ```
   rua clean -n .rua -n .cache -n .ignore -n compile_commands.json .
   ```

   ```
   bzhao in 🌐 build18-bj in ~/repos/MX_MAIN
   ❯ touch aaa
   bzhao in 🌐 build18-bj in ~/repos/MX_MAIN
   ❯ svn status .
   ?       .cache
   ?       .ignore
   ?       .rua
   ?       aaa
   ?       compile_commands.json
   M       sdk/phytium-kylin-sdk/linux/kernel/linux
   M       src/d-plane/flow/flow_first.c
   M       src/d-plane/policy/dp_policy.c
   bzhao in 🌐 build18-bj in ~/repos/MX_MAIN
   ❯ rua clean -n .cache -n .rua -n .ignore -n compile_commands.json
   [1/3] Removing target objs...ok
   [2/3] Removing WebUI objs...ok
   [3/3] Removing unversioneds...ok
   bzhao in 🌐 build18-bj in ~/repos/MX_MAIN
   ❯ svn status .
   ?       .cache
   ?       .ignore                    aaa 被清除
   ?       .rua
   ?       compile_commands.json
   M       sdk/phytium-kylin-sdk/linux/kernel/linux
   M       src/d-plane/flow/flow_first.c
   M       src/d-plane/policy/dp_policy.c
   ```
3. 在 ~/.rua/config.toml 或 $workspace/.rua/config.toml 中添加下面的配置内容，rua clean 在执行时会自动忽略这些文件：

   ```
   [clean]
   ignores = ["compile_commands.json", ".cache", ".rua", ".ignore"]
   ```

   ```
   bzhao in 🌐 build18-bj in ~/repos/MX
   ❯ svn status .
   ?       .cache
   ?       .ignore
   ?       .rua
   ?       aaa
   ?       compile_commands.json
   ?       src/mgmt/webserver/nginx/Makefile
   ?       target
   bzhao in 🌐 build18-bj in ~/repos/MX
   ❯ rua clean .
   [1/3] Removing target objs...ok
   [2/3] Removing WebUI objs...ok
   [3/3] Removing unversioneds...ok
   bzhao in 🌐 build18-bj in ~/repos/MX
   ❯ svn status .
   ?       .cache
   ?       .ignore
   ?       .rua
   ?       compile_commands.json
   bzhao in 🌐 build18-bj in ~/repos/MX
   ```

7 init

rua 可以在 bash/zsh/fish 等 shell 中自动补全。

7.1 用法

用户可以通过 rua init -h 查看帮助信息：

```
❯ rua init -h
Generate completion for the given shell

Usage: rua init <SHELL>

Arguments:
  <SHELL>  Shell type [possible values: bash, elvish, fish, powershell, zsh]

Options:
  -h, --help  Print help

Note:
  eval "$(rua init bash)"  # Append this line to ~/.bashrc
  eval "$(rua init zsh)"   # Append this line to ~/.zshrc
```

7.2 示例

1. 在 bash 中自动补全：

   ```
   eval "$(rua init bash)"
   ```
2. 在 zsh 中自动补全：

   ```
   eval "$(rua init zsh)"
   ```
3. 在 fish 中自动补全：

   ```
   eval (rua init fish)
   ```
4. 在 powershell 中自动补全：

   ```
   rua init powershell | Out-String | Invoke-Expression
   ```
5. 在 elvish 中自动补全：

   ```
   eval (rua init elvish)
   ```